

Ordinary multiplier norm detects finite Pick geometry

A qualitative operator-norm answer to Remark 4.3

Autonomous solve-first investigation, lane 12

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Abstract

Ofek–Pandey–Shalit ask whether their comparison of finite Drury–Arveson quotients and multiplier algebras remains valid with ordinary operator norm in place of completely bounded norm. We answer the qualitative form positively. For a fixed finite set X and arbitrary equal-cardinality sets Y_k , ordinary algebra distortion tending to one is equivalent to automorphism-invariant geometric convergence, RKHS Banach–Mazur convergence, and cb multiplier convergence. Consequently an ordinary-norm local modulus exists. The proof uses scalar multiplier extremals to prevent degeneration, compactness of normalized finite Gram matrices, continuity of multiplier norms, exact isometric rigidity, and Procrustes alignment. No explicit power-type modulus is obtained, so the source’s detailed numerical estimates are not all replaced.

1 Source question

The source is D. Ofek, S. K. Pandey, and O. M. Shalit, *Distance between reproducing kernel Hilbert spaces and geometry of finite sets in the unit ball*, J. Math. Anal. Appl. 500 (2021), 125140; arXiv:2011.06578v3 [1]. Its Remark 4.3 asks:

isomorphism $T : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ then one can say that $\rho_{RH}(\mathcal{H}_1, \mathcal{H}_2) = \infty$, but we will not deal with this case in this paper.

If $\varphi : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is a homomorphism between multiplier algebras, then we let $\|\varphi\|_{cb}$ denote the completely bounded norm of φ . This makes sense, because multiplier algebras are operator algebras (see Section 2.2).

Definition 4.2. Given two RKHSs $\mathcal{H}_i$ ($i = 1, 2$) with multiplier algebras $\mathcal{M}_i = \text{Mult}(\mathcal{H}_i)$ we define

$$\delta_M(\mathcal{M}_1, \mathcal{M}_2) = \inf \{ \|\varphi\|_{cb} \|\varphi^{-1}\|_{cb} : \varphi : \mathcal{M}_1 \rightarrow \mathcal{M}_2 \text{ a multiplier algebra isomorphism} \}.$$

The *multiplier Banach-Mazur distance* between $\mathcal{M}_1$ and $\mathcal{M}_2$ is defined to be

$$\rho_M(\mathcal{M}_1, \mathcal{M}_2) = \log(\delta_M(\mathcal{M}_1, \mathcal{M}_2)).$$

Again, this distance is always defined when $\mathcal{H}_1$ and $\mathcal{H}_2$ are of the same finite dimension. In the case of finite dimensional quotients of H_d^2 that we are interested in, every isomorphism is a multiplier algebra isomorphism (see Section 2.3), so we have in this case

$$\delta_M(\mathcal{M}_1, \mathcal{M}_2) = \inf \{ \|\varphi\|_{cb} \|\varphi^{-1}\|_{cb} : \varphi : \mathcal{M}_1 \rightarrow \mathcal{M}_2 \text{ is an algebra isomorphism} \}.$$

Remark 4.3. Some readers might wonder why we work with the completely bounded norm of the map φ and φ^{-1} rather than the operator norm of these maps, and whether we can obtain similar results if we work with the operator norm instead. The completely bounded norm is a natural choice for operator algebras, as we briefly discussed in Section 2.2; for a reader not familiar with this idea, we recommend [Pau02]. We do not know whether our results hold with the norm instead of the completely bounded norm.

The goal of this section is to study the connection between ρ_{RK} and ρ_M . We start with

For a finite set $X \subset \mathbb{B}_d$, write $\mathcal{H}_X = H_d^2|_X$ and $\mathcal{M}_X = \text{Mult}(\mathcal{H}_X)$. Define the ordinary multiplier distortion

$$\delta_{M, \text{op}}(\mathcal{M}_X, \mathcal{M}_Y) = \inf_{\varphi} \|\varphi\| \|\varphi^{-1}\|, \tag{1}$$

where $\varphi : \mathcal{M}_X \rightarrow \mathcal{M}_Y$ ranges over algebra isomorphisms. For finite sets, every isomorphism is composition with a bijection [1].

2 Proof intuition

Ordinary norm does not directly control the row amplification used in the source's cb proof. Instead, scalar multiplier extremals already recover each pseudohyperbolic distance. An ordinary near-isometry therefore prevents the moving point set from colliding or escaping to the boundary after one point is normalized to zero. Its finite Gram matrices are compact. At any Gram limit, continuity of multiplier norms turns “nearly isometric” into an exact isometric algebra isomorphism; exact rigidity then makes the limit congruent to the fixed set. Thus every Gram limit is the correct one, and a finite-dimensional Procrustes argument aligns the configurations.

3 Qualitative operator-norm theorem

Let $\tilde{\rho}_s$ denote the automorphism-invariant symmetric distance induced by ρ_{ph} , and let δ_{RK} and δ_M be the source's RKHS and cb multiplier distortions.

Theorem 1. *Fix $n \geq 2$ and a set $X = \{x_1, \dots, x_n\}$ of distinct points in the open unit ball of a complex Hilbert space. For any sequence $Y_k = \{y_1^k, \dots, y_n^k\}$ of n -point sets, the following are equivalent:*

1. $\tilde{\rho}_s(X, Y_k) \rightarrow 0$;
2. $\delta_{RK}(\mathcal{H}_X, \mathcal{H}_{Y_k}) \rightarrow 1$;
3. $\delta_M(\mathcal{M}_X, \mathcal{M}_{Y_k}) \rightarrow 1$;
4. $\delta_{M, \text{op}}(\mathcal{M}_X, \mathcal{M}_{Y_k}) \rightarrow 1$.

The assertion remains valid for an infinite-dimensional ambient Hilbert ball.

Corollary 2 (local modulus). *For every $\varepsilon > 0$ there is an $\eta > 0$, depending on X and ε , such that*

$$\delta_{M, \text{op}}(\mathcal{M}_X, \mathcal{M}_Y) < 1 + \eta \implies \tilde{\rho}_s(X, Y) < \varepsilon, \quad \delta_{RK}(\mathcal{H}_X, \mathcal{H}_Y) < 1 + \varepsilon, \quad \delta_M(\mathcal{M}_X, \mathcal{M}_Y) < 1 + \varepsilon$$

for every n -point set Y . In particular, ordinary near-isometry upgrades to complete near-isometry. Also, $\delta_{M, \text{op}}(\mathcal{M}_X, \mathcal{M}_Y) = 1$ if and only if X and Y are congruent.

4 Proof of the hard implication

We use the scalar extremal identity

$$\rho_{\text{ph}}(u, v) = \sup\{|f(u)| : \|f\|_{\mathcal{M}_Z} \leq 1, f(v) = 0\}, \quad u, v \in Z. \quad (2)$$

Indeed, contractive multipliers on a finite Drury–Arveson restriction extend contractively to the ball, giving “ $\leq$ ” by Schwarz–Pick; a scalar component of the ball automorphism taking v to zero attains equality.

Assume (4). Choose isomorphisms $\varphi_k : \mathcal{M}_X \rightarrow \mathcal{M}_{Y_k}$ induced by bijections and put $C_k = \|\varphi_k\|$, $D_k = \|\varphi_k^{-1}\|$, so that $C_k D_k \rightarrow 1$. The maps are unital, hence $C_k, D_k \geq 1$ and both tend to one. Label Y_k according to the bijections. Applying separate ball automorphisms to the matched first points preserves all relevant norms and distances, so assume $x_1 = y_1^k = 0$.

From (2),

$$C_k^{-1} \rho_{\text{ph}}(x_i, x_j) \leq \rho_{\text{ph}}(y_i^k, y_j^k) \leq D_k \rho_{\text{ph}}(x_i, x_j) \quad (1 \leq i, j \leq n). \quad (3)$$

Thus all y_i^k lie in a fixed smaller ball and remain uniformly separated. Let

$$G_k = (\langle y_i^k, y_j^k \rangle)_{i,j=1}^n.$$

Every subsequence has a further subsequence with $G_k \rightarrow G$. The positive semidefinite matrix G is the Gram matrix of a configuration $Z = \{z_1, \dots, z_n\}$ in a Hilbert ball; (3) makes its points distinct and $z_1 = 0$.

$$K_k = \left(\frac{1}{1 - \langle y_i^k, y_j^k \rangle} \right)_{i,j}$$

converge to the positive definite kernel matrix K_Z . If $a = (a_1, \dots, a_n) \in \mathbb{C}^n$, then $\|a\|_{\mathcal{M}_Z}$ is the operator norm of the diagonal map $k_{z_i} \mapsto \overline{a_i} k_{z_i}$ in the Hilbert norm with Gram matrix K_Z . Hence $K \mapsto \|a\|_{\mathcal{M}(K)}$ is continuous on positive definite matrices. On the other hand, the ordinary distortion gives, under the value-vector identifications,

$$D_k^{-1} \|a\|_{\mathcal{M}_X} \leq \|a\|_{\mathcal{M}_{Y_k}} \leq C_k \|a\|_{\mathcal{M}_X}.$$

Passing to the limit shows that the labeled identity $\mathcal{M}_X \rightarrow \mathcal{M}_Z$ is an isometric algebra isomorphism.

The exact rigidity theorem for Drury–Arveson multiplier varieties says that an isometric multiplier-algebra isomorphism is induced by a ball automorphism, and the varieties are congruent [2, 1]. Here the implementing automorphism respects the labels and sends $z_1 = 0$ to $x_1 = 0$; it is therefore unitary. Consequently $G = G_X$. Every convergent subsequence of (G_k) has the same limit, so $G_k \rightarrow G_X$.

Finally apply the Procrustes lemma from [1] in the span of the two n -tuples. That span has dimension at most $2n$, even if the ambient ball is infinite-dimensional. There are ambient unitaries U_k with

$$\max_i \|x_i - U_k y_i^k\| \rightarrow 0.$$

Euclidean and pseudohyperbolic metrics are uniformly equivalent on the fixed smaller ball, so $\tilde{\rho}_s(X, Y_k) \rightarrow 0$. This proves (4) $\Rightarrow$ (1).

The implications (1) $\Rightarrow$ (2) $\Rightarrow$ (3) are the quantitative results of [1], while (3) $\Rightarrow$ (4) follows from $\|\varphi\| \leq \|\varphi\|_{cb}$. Theorem 1 follows. Corollary 2 follows by contradiction: failure of any asserted modulus would give a sequence violating the theorem.

5 Scope and verification

This resolves the qualitative content of the source’s main finite-set equivalence with ordinary norms and supplies a local η - ε modulus. It does not produce the explicit power bounds in the source’s Proposition 4.7: ordinary norm does not directly control the vector-valued row multiplier used there.

The proof was audited at the scalar-extremal, nondegeneracy, kernel-norm continuity, labeled rigidity, and ambient-dimension steps. Exact-question, title/citation, ordinary-norm, multiplier Banach–Mazur, and later deformation searches on 13 August 2026 found later work using the cb formulation but no operator-norm answer. Novelty is plausible, not certified. Human review should focus on the labeled use of exact rigidity and on whether the source intended an explicit quantitative replacement rather than its qualitative main theorem.

References

- [1] D. Ofek, S. K. Pandey, and O. M. Shalit, *Distance between reproducing kernel Hilbert spaces and geometry of finite sets in the unit ball*, J. Math. Anal. Appl. 500 (2021), 125140; arXiv:2011.06578.
- [2] K. R. Davidson, C. Ramsey, and O. M. Shalit, *Operator algebras for analytic varieties*, Trans. Amer. Math. Soc. 367 (2015), 1121–1150.